Fernando Bellido Pazos

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WORK EXPERIENCE

DXC Technology Zaragoza, Spain
AI Engineer & Technical Leader Jan. 2025 – Today

Description:

1. **Enterprise AI Platform Leadership**
    - Led cross-functional teams to design and launch a **Microsoft AutoGen-powered** multi-agent AI platform, managing various microservices and cloud-native architecture on Azure.
    - **Architected intelligent agents** (geospatial, claims, decision automation) at enterprise scale using Azure OpenAI GPT-4o.
    - Designed **event-driven data pipelines** (Azure EventHub, 99.9% reliability), processing over 1M records.
    - Owned **end-to-end delivery** from technical roadmap to enterprise deployment.
    - Built DevOps pipelines, **CI/CD automation, and AI model validation** frameworks for scalable production systems.
    - **Mentored teams on advanced AI**, multi-agent orchestration, and secure execution environments.
2. **Agentic Traffic Management Automation Platform:**
    - Designed and implemented a multi-agent geospatial planning engine to automate temporary traffic management layouts from high-level roadwork requests, **using Google ADK, Agent2Agent, MCP, PostgreSQL, Azure and Kubernetes**.
    - Built an orchestration layer coordinating **specialised agents and shared MCP tools**, ensuring deterministic, auditable workflows and **clean separation between business logic, standards and geospatial computation**.
    - Integrated the platform with existing enterprise systems (including CRM and work management tools) via dedicated adapters, enabling **end-to-end automation from request intake to deployment-ready** traffic plans.
    - Improved **delivery speed and consistency of traffic management designs** by standardising computations, reducing manual configuration effort and lowering the risk of human error in layout definition.
3. **Contract Intelligence Platform**
    - Built production-grade conversational AI with **Google ADK & RAG**, enabling 10K+ contract searches with >90% retrieval accuracy.
    - Developed **concurrent AI pipelines** for contract term extraction via Azure OpenAI GPT — cutting processing time by 70%.
    - Engineered microservice architecture with **Model Context Protocol (MCP), vector search (Azure AI Search),** and real-time streaming APIs.
    - Delivered **full-stack platform** (FastAPI, PostgreSQL, Azure AKS, Azure AD) with robust cloud infrastructure and auto-scaling Kubernetes deployments.
    - Automated legal compliance reviews, **reducing review cycles** from days to minutes.

IDOM | Consulting, Engineering, Architecture Bilbao, Spain
Consultant & System Innovator Sept. 2023 – Dec. 2024

Description:

1. **System Innovator:**
    - Developed a Proof-of-Concept **Digital Twin** utilizing **Autodesk Tandem** and **Azure IoT Hub**, in conjunction with **Azure Stream Analytics** and **Azure ML Studio**, to showcase advanced capabilities. Demonstrated that this approach could outperform traditional human monitoring systems by providing real-time data integration and predictive analytics.
    - Directed the theoretical research & development of an innovative internal **Retrieval Augmented Generation (RAG)** system, significantly enhancing language understanding capabilities and applications within the organization. Spearheaded the creation of an AI-driven companion, demonstrating leadership in leveraging advanced AI technologies & Large Language Models (LLM) to drive organizational progress and innovation.
    - Pioneered the development of an **AI classifier using NLP and the Google's BERT** language model, complemented by an ontology-based approach, for sophisticated categorization derived from comprehensive project descriptions.
    - Innovated a **CV generator** grounded in the S&P's Success Factors internal system, featuring a multi-layered approach encompassing data ingestion, AI-driven & deterministic data processing paragraph generation, template compilation, and polished CV output.
    - Architected and deployed a cutting-edge AI development infrastructure, encapsulating a fully **dockerized** framework that enables real-time monitoring of system resources. Masterfully set-up **Traefik** for efficient routing, and implemented a suite of advanced AI tools, setting new benchmarks in technological excellence like **Stable Diffusion**.
2. **Consulting – Digital Transformation:**
    - Developed an **AI-automated pipeline** for processing tabular data into specific SAP naming conventions using **Azure Document Intelligence** and **Azure Phi-3.5** cloud deployment. This system significantly reduced data processing time and improved the accuracy and consistency of data integration within SAP systems compared to manual human processing.
    - Spearheaded the development of cutting-edge **artificial intelligence and mathematical models**, revolutionizing asset maintenance from reactive to proactive using S&P's Intelligent Asset Management and Internet of Things modules.
    - Played a pivotal role as a Statistician in developing the cadastre system, in collaboration with the registry office, offering clear justifications for property valuations to citizens. Innovated a **unique calculation method for assessing house values** across the Basque Country, enhancing the system's accuracy and transparency in property valuation.

Ontology Engineering Group Madrid, Spain
Data Analyst and Developer Intern Sept. 2021 – Sept. 2022
Description:

- Developed a sophisticated system to construct **knowledge graphs from academic papers** housed within institutional repositories, enhancing advanced querying capabilities related to Open Science principles.
- Analyzed large datasets of academic publications to extract meaningful metadata and relationships, improving data accessibility and facilitating deeper insights for researchers.
- Implemented **ontology-based data models** to represent complex scholarly information, enabling better interoperability and integration with other data sources.
- **Optimized data ingestion** processes by automating metadata extraction and semantic annotation, increasing efficiency by 40% compared to manual methods.
- Collaborated with a multidisciplinary team to refine algorithms for data processing and knowledge representation, contributing to the advancement of open-source tools in the academic community.
- Utilized technologies such as **RDF**, **SPARQL**, and Python libraries for **semantic data processing**, enhancing the group's capabilities in semantic web technologies.

Savestars Consulting Madrid, Spain
Developer June 2020 – July 2021
Description:

- Developed a Python-based system to extract images from **NASA's VNP46A2** satellite data, enabling **detailed monitoring** and **analysis of nocturnal light emissions**.
- Processed satellite images and applied geospatial masking techniques to focus on designated areas of the peninsula, providing precise assessments of light pollution levels in specific regions.
- Engineered algorithms for **image correction** and **enhancement**, improving the accuracy of light intensity measurements by 25%.
- Conducted **temporal data analysis** to identify patterns and trends in light pollution over time, supporting environmental research and policy-making.
- **Collaborated** with **environmental scientists** to interpret data results, contributing to initiatives aimed at reducing light pollution and its ecological impacts.
- Utilized tools such as **GDAL, NumPy**, and **Matplotlib** for geospatial data processing and visualization, advancing the firm's technical expertise in environmental data analysis.

EDUCATION

Carlos III University of Madrid Madrid, Spain
MSc in Applied & Computational Mathematics [English] Sept. 2022 - June 2023
GPA: 8.38/10 with 1 Honourable Mention in Advanced Analysis.
Relevant Coursework: Biological Inspired Artificial Intelligence, Modelling and Nonlinear Analysis, Optimization, Stochastic Equations for Finance and Biology, Complex Biological and Socioeconomic systems, Computational Linear Algebra, Computational techniques for Differential equations, Methods for Nonlinear differential equations, Applied discrete mathematics.

Madrid's Polytechnic University Madrid, Spain
Bachelor's degree in mathematics & computer science [Spanish] Sept. 2018 - June 2022
GPA: 9.4/10 with 14 Honourable mentions.
Relevant Coursework: Data Analysis, Software Engineering; Operating Systems, Artificial Intelligence, Databases, Modelling, Programming (Imperative, Object Oriented, Functional, Logic, Systems), IT Project Management, Topology, Calculus, Algebra, Pure Mathematics.

PROJECTS

Basic Numerical Methods in Python - Student Oriented June 2023
Description: My end of master's thesis aims to develop an open-source PyPi project that includes a scripted version of the numerical methods taught in universities and a Jupyter-integrated GUI for real-time visualization of those. The goal is to help students understand the methods and promote learning Python instead of MATLAB, which is not as popular or accessible as Python in the current job market.

- This project was awarded the 'Most Innovative Project of 2023' by Logicalis in the Logicalis Innovation Challenge, recognizing its potential to significantly impact educational practices and student learning experiences.

Sturm Liouville Problem and Models based on differential equations on the field of Acoustics June 2022
Description: Pursued a comprehensive study of the Sturm-Liouville theory and modelled stringed musical instruments to explore the relationship between mathematics and music. The aim of the research was to answer questions such as: what factors contribute to the distinct sound of instruments like a violin and a piano, and how does mathematics play a role in the creation of music.

AWARDS

- Logicalis Innovation Challenge
- Cambridge Advanced Certificate (C1)
- Academic Excellence Grant 2021 & 2022
- Academic Excellence Award for Representative Agents of Banco Mediolanum 2022

ADDITIONAL

People Skills: Leadership, Public Speaking, Critical Thinking, High Self-Motivation, Dependable
Languages: English (Native) and Spanish (Native)